Weak solution of the merged mathematical equations of the polluted atmosphere

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Abstract

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Keywords: Navier-Stokes equations, shallow domains, pollution evolution equation, hydrostatic approximation.

Contents

1 Introduction

We investigate existence of a weak solution for the merged motion-pollution atmosphere model. At this point the functions we work with are $\mathbf{v} = (\mathbf{v}_H, v_3)$, and C, representing velocity and concentration, respectively (we are not considering other variables such as water vapor quantity or temperature).

We underline the fact that this work is for the case of **inland** scenario.

Contaminants emitted into the atmosphere are transported by the wind, turbulent behaviour results in mixing and dispersion, gravity acts to deposit containment on the ground.

2 Equations describing the polluted atmosphere

The first step is to choose a coordinate system, set the domain we start with and an appropriate set of equations describing both atmosphere motions and the effect of pollution. The set of equations, the choice of the locally acceptable Cartesian coordinate system we are using for the velocity terms are similar to those suggested by the [Azérad and Guillén, 2001] article, we ignore large scale effects such as the curvature of the Earth. For describing the effect of pollution we use a continuity equation written in Cartesian coordinates for the concentration.

The coordinate system we choose to work with is local, we neglect curvature and work with Cartesian coordinates which is chosen to be adjusted to the downwind direction, ie. z is the vertical direction, while x is the downwind, y is the crosswind direction.

The original domain:

$$\Omega_\epsilon = \tag{1}$$

The equations describing the atmosphere are the incompressible Navier-Stokes:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0, T) \tag{2}$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0, T) \tag{3}$$

dispersion of pollutants is commonly modeled by convection / advection - diffusion equation. The diffusion-reaction equation describing the pollutant chemicals in the air:

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2} + S \tag{4}$$

3 Rescaling

3.1 Rescaling due to shallow domain

$$x = x_1, y = x_2, z = \epsilon x_3$$

$$v_1 = u_1^\epsilon, v_2 = u_2^\epsilon, v_3 = \epsilon u_3^\epsilon, p = p^\epsilon$$
$$\nu_x = \nu_1, \nu_y = \nu_2, \nu_z = \epsilon^2 \nu_3$$
$$\tau_i^\epsilon = \epsilon \theta_i$$
$$K_z = \epsilon^2 K_3$$

So we have a new domain

$$\Omega =$$

3.2 Rescaling due to a special choice of coordinate system

Here we exploit the fact that we chose the coordinate system in a way so that the x axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, ie.

$$|v_1 \partial_x C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right| \tag{5}$$

which leads us to using
$$K_x = \epsilon K_1$$

Putting this condition and the previously described rescaling terms together to the original equations, we arrive to the merged anisotropic equations of the polluted atmosphere above inland surface.

3.3 Rescaled merged equations - Anisotropic Navier-Stokes plus pollution

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{6}$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{7}$$

$$\epsilon^2 \{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{8}$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{9}$$

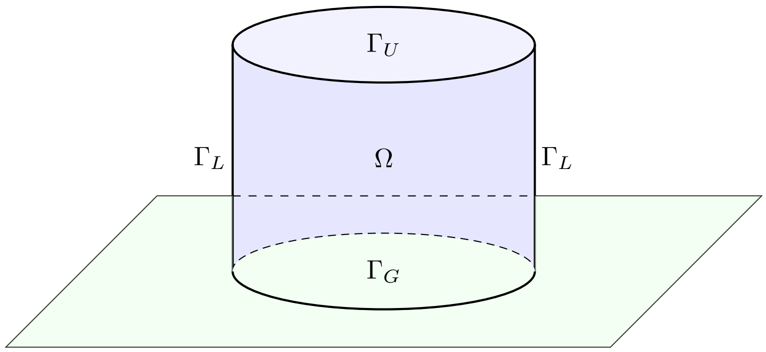
$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0,T) \tag{10}$$

$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, C^\epsilon(\cdot, t=0) = C_0^\epsilon \text{ in } \Omega \tag{11}$$

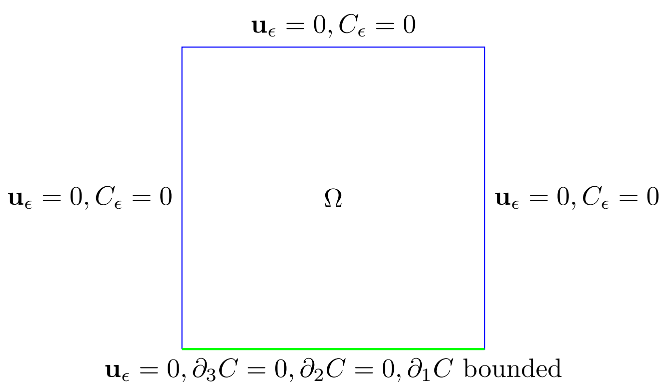
$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S \tag{12}$$

4 Boundary conditions

It is vital to choose boundary conditions for the pollution concentration that are physically as accurate as possible and at the same time not unnecessarily complicated.



The boundary division of the $\partial\Omega$: the upper boundary of the Ω domain is denoted by Γ_U, the lateral boundary is Γ_L, and the lower boundary of the domain is Γ_G. As we mentioned before, the case we are investigating is that of inland domains Ω, ie the case of landscapes and cities with no significant nearby water surface. This is very important since the boundary conditions for both wind velocity and pollution concentration in the air air different depending whether we are above solid ground or water surface. The boundary conditions we choose to use in the inland scenario are the following, shown in a cross-section.



An explanation of the boundary conditions:

- The upper boundary of the domain we are considering is chosen to be at the isobar representing the planetary boundary layer ([Arya, 2001]), on this $p = p_0$ pressure level the upper boundary of the local domain is denoted by Γ_U. On Γ_U we use $\mathbf{u}_H = 0, u_3 = 0, C(x,y,z) = 0$ boundary conditions.

- On both sides the lateral boundary of the rectangle shaped Ω domain is given by Γ_L. Here we use the same boundary conditions as on the upper level, ie. we have $\mathbf{u}_H = 0, u_3 = 0, C(x,y,z) = 0$.

  Remark. *If we wanted to use $C(x,y,z) = c$ with some c constant value, we could make a change of variables distracting that background flow and arrive back to the same mathematical problem with C being zero.*

- On the $z = 0$ ground level we have $u_H = 0, u_3 = 0$ boundary conditions for the air velocity. This is the mathematical way of expressing no-slip boundary conditions for the horizontal wind speed and the impervious nature of the ground with respect to the wind. As for the concentration, the way we can choose ground level boundary conditions is the following. Firstly, we assume that the pollution is not absorbed by the ground, so we use $\partial_3 C = 0$ on Γ_G (see also for example [Yeh and Huang, 1975]). For the horizontal partial derivatives of the pollution, we can separate downwind and crosswind direction boundary conditions. In the crosswind direction we use $\partial_2 C = 0$ while in the downwind (stronger) direction our assumption is more relaxed, we only assume that $\partial_1 C$ is finite in the boundary norm, ie in $L^2L^2(0,T;\Gamma_G)$.

Remark. *In the paper of [Temam and Ziane, 2005], on p35 it is suggested that the domain we work on has the form of $p_0 < p < p_1$, because for the thin portion of air between the earth and the isobar $p = p_1$, another specific simplified model would be necessary. So a way to make the model more precise is to cut the domain at an isobar a little bit off the physical ground. This would of course radically change the boundary conditions we use for the functions, because it would take away the advantage of having a wall-like, natural and physical boundary condition for our domain to support it. If we are positively above the ground, $\partial_3 C = 0$ would represent a strange reflective layer.*

5 Hydrostatic Navier-Stokes plus pollution

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0,T) \tag{13}$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0,T) \tag{14}$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0,T) \tag{15}$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0,T) \tag{16}$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0,T) \tag{17}$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1{,}2 \tag{18}$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \tag{19}$$

6 Main theorem

- The article of [Azérad and Guillén, 2001] proved an existence theorem for weak solution of the equations describing the atmosphere without pollution.

  Theorem 1. *Let* $\mathbf{u}_0 \in L^2(\Omega)^3$, *with* $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ *on* $\partial\Omega$, *and* $\theta_1, \theta_2 \in L^2(0, T; H^(- 1/2)(\Gamma_s))$; *there exists a weak solution* $\mathbf{u}$ *of the hydrostatic Navier-Stokes equations as the aspect ratio* ϵ *tends to* 0.

- The task we want to do is to extend this result in the sense that it becomes valid to the merged set of equations as well, $\mathbf{s} = (\mathbf{u}, C)$.

7 Weak formulation

Weak formulation of the merged equations in both the anisotropic and hydrostatic case.

7.1 Weak formulation of the anisotropic system

$$\begin{gathered}
\int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\
+ \epsilon\beta \big[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3) \big] dt \\
+ \epsilon^2 \big[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3) dt \big] \\
+ \int_0^T -(C^\epsilon, \partial_t \tilde{C}^\epsilon) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon) + \\
\epsilon K_1 (\partial_1 C^\epsilon, \partial_1 \tilde{C}^\epsilon) - \epsilon K_1 \big[(\partial_1 C^\epsilon) \tilde{C} \big]_{\Gamma_G} + K_2 (\partial_2 C^\epsilon, \partial_2 \tilde{C}^\epsilon) + K_3 (\partial_3 C^\epsilon, \partial_3 \tilde{C}^\epsilon) dt = \\
= -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H^\epsilon(0)) - \epsilon^2 (u_{03}^\epsilon, v_3^\epsilon(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{gathered} \tag{20}$$

To pass to the limit in the terms that do not include pollution in the anisotropic equation has already been worked out in the [Azérad and Guillén, 2001] article. What we need to work on is how to legitimately pass to the limit with the C terms. To do this we firstly will use the key tool of the previously mentioned article, the energy inequality. Once we have a priori estimates for the functions in the linear terms, we can pass to the limit with those without particular problems; the challenging part will be to prove convergence for the $(\mathbf{u}^\epsilon C^\epsilon, \nabla\tilde{C}^\epsilon)$ term, here we will need a stronger result than just the weak convergence of the present functions in the product.

7.2 Weak formulation of the hydrostatic system

Add specific assumptions on the test functions.

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u}\cdot\nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H) dt \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla\tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C}) dt = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned} \tag{21}$$

8 Energy inequality and a priori estimates

8.1 Energy inequality

Calculating the energy inequality for the merged system we arrive to

$$\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(t)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(t)\right\|_{L^2}^2 + \left\|C^\epsilon(t)\right\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
&\int_0^t \{\epsilon K_1\|\partial_1 C^\epsilon\|_{L^2}^2 + K_2\|\partial_2 C^\epsilon\|_{L^2}^2 + K_3\|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \\
&\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(0)\right\|_{L^2}^2 + \epsilon^2\left\|u_3^\epsilon(0)\right\|_{L^2}^2 + \left\|C^\epsilon(0)\right\|_{L^2}^2) + \int_0^t (S, C) d\tau + \int_0^t \epsilon K_1 \langle \partial_1 C, C\rangle_{\Gamma_G} d\tau
\end{aligned} \tag{22}$$

In order to use the energy inequality to extract a priori estimates, we need to know that the right hand side of the inequality is bounded. Since the S source term is assumed to be as smooth as we need it to be, the only term we need to work further is $\int_0^t \epsilon K_1 \langle \partial_1 C, C\rangle_{\Gamma_G} d\tau$.

We can estimate this term in the following way.

$$\begin{aligned}
&\int_0^t \epsilon K_1 \langle \partial_1 C, C\rangle_{\Gamma_G} d\tau \leq K_1 \int_0^t \sqrt{\epsilon}\|\partial_1 C\|_{L^2(\Gamma_G)} \sqrt{\epsilon}\|C\|_{L^2(\Gamma_G)} \leq \\
&\frac{1}{\delta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C\|_{L^2(\Gamma_G)}^2 + \delta K_1 \int_0^t \frac{\epsilon}{2}\|C\|_{L^2(\Gamma_G)}^2 \leq \\
&B.T. + \delta\frac{\epsilon}{2} K_1 c_T \int_0^t \|C\|_{W^{1,2}(\Omega)}^2 ,
\end{aligned} \tag{23}$$

where c_T is the constant provided by the trace theorem, and $B.T.$ denotes the boundary term which is bounded because of the a priori assumptions made on boundary conditions. Now we can carry on with the estimate by using the $W^{1,2}$ norm definition, exploit the fact that we are on a bounded domain and then applying the Sobolev inequality (for $n = 3, p = 2, p* = 6$, denoting its constant by c_S), so we have

$$\begin{aligned}
&\int_0^t \epsilon K_1 \langle \partial_1 C, C\rangle_{\Gamma_G} d\tau \leq \\
&B.T. + \delta\frac{\epsilon}{2}K_1 c_T \int_0^t \{\|C\|^2_{L^2(\Omega)} + \|\nabla C\|^2_{L^2(\Omega)}\} d\tau \leq \\
&B.T. + \delta\frac{\epsilon}{2}K_1 c_T \int_0^t \{\|C\|^2_{L^6(\Omega)} + \|\nabla C\|^2_{L^2(\Omega)}\} d\tau \leq \\
&B.T. + \delta\frac{\epsilon}{2}K_1 c_T \int_0^t \{c_S\|\nabla C\|^2_{L^2(\Omega)} + \|\nabla C\|^2_{L^2(\Omega)}\} d\tau = \\
&B.T. + \delta\frac{\epsilon}{2}K_1 c_T (c_S + 1) \int_0^t \|\nabla C\|^2_{L^2(\Omega)}\, d\tau
\end{aligned} \tag{24}$$

Which means that the energy inequality takes the form of

$$\begin{aligned}
&\frac{1}{2}(\left\|\mathbf{u}^\epsilon_H(t)\right\|^2_{L^2} + \epsilon^2\left\|u^\epsilon_3(t)\right\|^2_{L^2} + \left\|C^\epsilon(t)\right\|^2_{L^2}) + \int_0^t \{\|\nabla_\nu \mathbf{u}^\epsilon_H\|^2_{L^2} + \epsilon^2\|\nabla_\nu u^\epsilon_3\|^2_{L^2}\} d\tau + \\
&\int_0^t \{(\epsilon K_1 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_1 C^\epsilon\|^2_{L^2} + (K_2 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_2 C^\epsilon\|^2_{L^2} + \\
&+ (K_3 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_3 C^\epsilon\|^2_{L^2}\} d\tau \leq \\
&\frac{1}{2}(\left\|\mathbf{u}^\epsilon_H(0)\right\|^2_{L^2} + \epsilon^2\left\|u^\epsilon_3(0)\right\|^2_{L^2} + \left\|C^\epsilon(0)\right\|^2_{L^2}) + \int_0^t (S, C) d\tau + \frac{1}{\delta}K_1\frac{\epsilon}{2}\int_0^t \|\partial_1 C\|^2_{L^2(\Gamma_G)}
\end{aligned} \tag{25}$$

The right hand side of the energy inequality is now apparently bounded, and it is easy to see that the coefficients of the pollution terms are positive, since the δ constant of the generalized Young inequality can be freely chosen in a way such that the requirements $\epsilon K_1(1 - \delta\frac{1}{2}c_T(c_S+1)) > 0, (K_2 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0, (K_3 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0$ are all satisfied.

8.2 A priori estimates and weak convergence

Now we can extract a priori estimates from the final form of the energy inequality. We have

- $L^\infty_t L^2_x$ boundedness for $u_1, u_2, \epsilon u_3$,
- $L^2_t H^1_x$ boundedness for $u_1, u_2, \epsilon u_3$,

- $L_t^\infty L_x^2$ boundedness for C,
- $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon}\partial_1 C, \partial_2 C, \partial_3 C$.

Because of the boundedness we have weak convergence, ie. there are subsequences still denoted by the same way, for which we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ and $\partial_3 C^\epsilon \rightharpoonup \partial_3 C$ in $L_t^2 L_x^2$, and $\sqrt{\epsilon}\partial_1 C^\epsilon \rightharpoonup e$ for some limit element e in $L_t^2 L_x^2$.

9 Passing to the limit

Passing to the limit is done without problems for the terms that do not include pollution term, since that has been proved before in the [Azérad and Guillén, 2001] article. In the linear terms the weak convergence guaranteed by the a priori estimates will be enough. In the nonlinear term we need an additional result that provides a weak convergence of the product of functions for which we only had weak convergence separately.

9.1 Linear terms

For the linear terms we can directly apply the weak convergence results of the previous section.

(i) The convergence of the $(C^\epsilon, \frac{\partial \tilde{C}}{\partial t})$ term is easy to see, since we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$.

(ii) For the term we have $\epsilon K_1 \int_0^T (\partial_1 C^\epsilon, \partial_1 \tilde{C}) = \sqrt{\epsilon} K_1 \int_0^T (\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_1 \tilde{C})$, which goes to 0 as $\epsilon \to 0$ because of the Cauchy-Schwartz inequality, since we have $\sqrt{\epsilon}\partial_1 C^\epsilon$ bounded in $L_t^2 L_x^2$, and $\tilde{C}$ is a test function.

(iii) As we have seen before, we have $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ in $L_t^2 L_x^2$, which by definition provides that

$$\int_0^T (\partial_2 C^\epsilon, f) \mathrm{d}t \to \int_0^T (\partial_2 C, f) \mathrm{d}t \qquad \forall f \in L_t^2 L_x^2.$$

(iv) Analogously,

$$\int_0^T (\partial_3 C^\epsilon, f) \mathrm{d}t \to \int_0^T (\partial_3 C, f) \mathrm{d}t \qquad \forall f \in L_t^2 L_x^2.$$

(v) The $\epsilon K_1 \int_0^T \left[(\partial_1 C^\epsilon) \tilde{C} \right]_{\Gamma_G} \mathrm{d}t$ line integral vanishes as $\epsilon \to 0$ since by the Cauchy-Schwartz equation we can estimate the integral by two bounded terms ($\partial_1 C^\epsilon$ was assumed to be finite in norm on the lower boundary).

9.2 Nonlinear term

We need some sort of compactness theorem to handle $(\mathbf{u}^\epsilon C^\epsilon, \nabla\tilde{C})$.

Remark. *As a matter of fact, we actually have three terms here, namely $(u_i^\epsilon C^\epsilon, \partial_i\tilde{C})$ for $i = 1, 2, 3$, and for the values $1, 2$ of i the weak convergence of the $u_i^\epsilon C^\epsilon$ product is given by the fact that, as it is proved in the reference article, we actually have strong convergence for the horizontal velocities, and because of the a priori estimate from the energy bound we know we have weak convergence for the C^ϵ term. But this argument does not really solve the challenge caused by nonlinearity because it does not hold for u_3, so we still need an additional compactness theorem as mentioned before.*

In this case the Aubin-Lions lemma can not be applied since it would require for C^ϵ to be in a compactly embedded space such as L^2H^1, which is not the case here because we only know boundedness results for $\sqrt{\epsilon}\partial_1 C$, but not for $\partial_1 C$.

We will show that the following lemma from [Lions, 1996] is applicable in our case.

Lemma 1. *Let g_n, h_n converge weakly to g, h respectively in $L^{p_1}(0,T;L^{p_2}(\Omega))$, $L^{q_1}(0,T;L^{q_2}(\Omega))$ where $1 \le p_1, p_2 \le +\infty$,*

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2} = 1.$$

We assume in addition that

- *$\frac{\partial g^n}{\partial t}$ is bounded in $L^1(0,T;W^{-m,1}(\Omega))$ for some $m \ge 0$ independent of n.*
- *$\left\|h^n - h^n(\cdot + \xi, t)\right\|_{L^{q_1}(0,T;L^{q_2}(\Omega))} \to 0$ as $|\xi| \to 0$, uniformly in n.*

Then, $g^n h^n$ converges to gh (in the sense of distributions on $\Omega \times (0,T)$).

What we will show in the following is that for the functions u_i^ϵ and C^ϵ the assumptions of Lemma 1 are satisfied, which means that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ holds.

Lemma 2. *We have $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$.*

Proof. We have seen from the a priori estimates that $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, and we know from the reference article that $u_i^\epsilon \rightharpoonup u_i$ in $L_t^2 L_x^2$.

Let's choose $q_1 = q_2 = p_1 = p_2 = 2$. This is fine since we are on a finite domain, so $C^\epsilon \in L^\infty L^2$ implies $C^\epsilon \in L^2 L^2$.

- **Bound for the time derivative**.

$$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon}\, \underbrace{K_1\partial_1 \underbrace{\partial_1\sqrt{\epsilon}C^\epsilon}_{L^2}}_{W^{-1,2}} + \underbrace{K_2\partial_2 \underbrace{\partial_2 C^\epsilon}_{L^2}}_{W^{-1,2}} + \underbrace{K_3\partial_3 \underbrace{\partial_3 C^\epsilon}_{L^2}}_{W^{-1,2}} - S$$

Since the S source term was assumed to be very regular, we see that the second part of the right hand side is in $W^{-1,2}$. On the other hand, the first part is in $W^{-1,1}$, because we have

$$u^\epsilon \nabla C^\epsilon = \underbrace{\nabla \cdot (\underbrace{u^\epsilon C^\epsilon}_{L^1})}_{W^{-1,1}} - \underbrace{\nabla \cdot (u^\epsilon)}_{0} C.$$

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$.

- **Equicontinuity**. We need to show that $\left\|u_i^n - u_i^n(\cdot + \xi, t)\right\|_{L^2 L^2} \to 0$ as $\xi \to 0$. Since in the time domain the condition naturally holds, we will focus on the space part of the proof.

  First we show that for a function f in the function space C_0^∞ the condition holds.

$$\left\|f(x+\xi) - f(x)\right\|_{L^2} = (\int_\Omega \left|f(x+\xi) - f(x)\right|^2 dx)^{1/2} \leq$$
$$\left(\int_\Omega (\sup_x \left|f(x+\xi) - f(x)\right|)^2 d\mathrm{x}\right)^{1/2} = \left\|f(x+\xi) - f(x)\right\|_{L^\infty} \cdot |\Omega|^{1/p}.$$

  So we have

$$\lim_{\xi \to 0} \left\|f(x+\xi) - f(x)\right\|_{L^2} \leq |\Omega|^{1/p} \cdot \lim_{\xi \to 0} (\sup_x \left|f(x+\xi) - f(x)\right|) = 0,$$

  since $f \in C_0^\infty$.

  As a second step we show that the condition holds for any function $f \in L^2$. Here we exploit the fact that C_0^∞ is dense in the L^p function space. By definition this means that $\exists \varphi_n \in C_0^\infty(\Omega)$ such that $\varphi_n \to f$ in $L^p(\Omega)$. With this we have

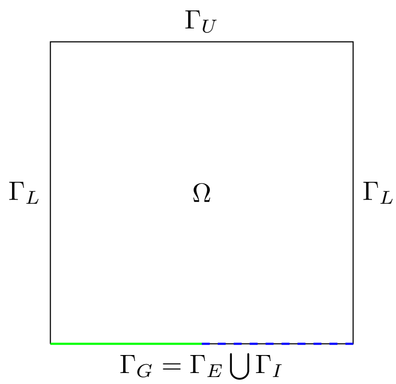
$$\left\|f(x+\xi) - f(x)\right\|_{L^2} =$$
$$\left\|f(x+\xi) - \varphi_n(x+\xi) + \varphi_n(x+\xi) - \varphi_n(x) + \varphi_n(x) - f(x)\right\|_{L^2} \leq$$
$$\left\|f(x+\xi) - \varphi_n(x+\xi)\right\|_{L^2} + \left\|\varphi_n(x+\xi) - \varphi_n(x)\right\|_{L^2} + \left\|\varphi_n(x) - f(x)\right\|_{L^2}$$

  The first and third term goes to 0 as $\xi \to 0$, since $\varphi_n \to f$ in $L^p(\Omega)$. On the other hand in the first part of the proof we have shown exactly that the second term goes to 0. Which proves that $\left\|u_i^n - u_i^n(\cdot + \xi, t)\right\|_{L^2 L^2} \to 0$ as $\xi \to 0$. Eventually this means that we can apply the lemma with $h^n = u_i^\epsilon, g^n = C^\epsilon$, which proves the statement of the lemma.

$\square$

10 Future work

- Use more sophisticated ground-level boundary conditions instead of $\partial_3 C = 0$ in the inland case, for example [Monin, 1959].
- Making the model physically more precise, accurate, and adapted for near seaside conditions are the following.



On the ground layer we naturally separate the boundary into two different parts, Γ_E denoting the part of the boundary above the ground, and Γ_I denoting the section that is interacting with the sea. The separation is a sort of expansion or generalization of the ground boundary since while above ground parts it is reasonable to use no-slip conditions, above the ocean it is natural to assume that the wind has an interaction with the waves, there is no such rigidness as in the ground case. For this case we could use $\nu_3 \frac{\partial \mathbf{u}_H}{\partial x_3} = \theta_H, u_3 = 0$. For the pollution concentration as well, above sea territory it is not natural to use $\partial_3 C = 0$.

11 TODO

1.) Write down the whole process for the version when we touch the ground and there is no ocean. Specify spaces.

2.) When we touch the ground and there is ocean but no eps in the continuity eq

3.) Look into the physics books

4.) Look into the 2 articles and the idea boundary condition which could be a way for improvement.

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